

The Lean formalization of Gwynne–Miller’s LQG metric theorems: differences from the papers

3 October 2026

1 Scope

The project is a Lean 4/mathlib formalization of Theorems 1.1 and 1.2 of Gwynne–Miller, *Existence and uniqueness of the Liouville quantum gravity metric for $\gamma \in (0, 2)$* (arXiv:1905.00383, “GM”). The final theorems are

```
LQGMetric.theorem11_proved : Theorem11, theorem12_proved : Theorem12,  
main_result
```

in `LQGMetric/Assembly/MainFinal.lean`; the third bundles both theorems with the existence of a normalized whole-plane GFF. They have no hypotheses and depend only on Lean’s three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). They were kernel-checked twice: a `leanchecker --fresh` replay of the whole import closure including `mathlib`, and a per-module replay of all 3399 project modules. The library has 2361 files and about 382,000 lines.

Every result that GM’s proof uses from the cited papers is proved in Lean: CONF (Gwynne–Miller, confluence of geodesics, arXiv:1905.00381), DG (Ding–Gwynne, 1807.01072), DZZ (Ding–Zeitouni–Zhang, 1807.00422), DDDF (Ding–Dubédat–Dunlap–Falconet, 1904.08021), DFGPS (Dubédat–Falconet–Gwynne–Pfeffer–Sun, 1905.00380), LM (Gwynne–Miller, local metrics, 1905.00379), MQ (Miller–Qian, 1812.03913) and DGo (Ding–Goswami, 1610.09998). Only what GM use is proved, not the whole papers.

“Differs” is taken at three levels: the statements of Theorems 1.1 and 1.2 and the conventions behind them (§2); cited results proved in a different form (§3); proofs that follow a different route (§4). §5 lists conventions that run through everything, §6 says where the details are. Nothing in §3–§5 touches the statements of §2: every departure there is a kernel-checked lemma inside the proof, not an assumption.

2 The statements of Theorems 1.1 and 1.2

The statement layer (eight files under `LQGMetric/Statement/`) has been frozen since 1 October 2026, after an independent audit, with the readings S1–S7 (statements) and F1–F13 (definitions) listed in `STATEMENT_SPEC.md`. It was later re-audited against GM (l. 202–231, 255–322 of the LaTeX source) and against the original sources of each definition (Duplantier–Sheffield, DZZ, DG, DDDF, DFGPS, LM). No meaning-level mismatch was found; the one source-level finding is the normalization of $\mathfrak{a}_\varepsilon$ below. The conventions are:

- **Distributions.** $\mathcal{D}'(\mathbb{C})$ is `mathlib`’s space of continuous linear functionals on real C_c^∞ test functions, with the cylinder σ -algebra generated by the pairings $h \mapsto \langle h, \varphi \rangle$ (GM do not name one; it coincides with the Borel σ -algebra of the weak- $*$ topology). “Measurable $h \mapsto D_h$ ” refers to it.
- **Whole-plane GFF.** A random h is a whole-plane GFF if it is measurable and its pairings with mean-zero test functions form a centered Gaussian process with covariance $\iint \varphi(x)(-\log|x-y|)\psi(y) dx dy$. The additive constant is unconstrained and may be random; this is GM’s “defined

modulo a global additive constant”. The circle average $h_r(z)$ is the limit of the circle averages of mollified fields; “normalized” means $h_1(0) = 0$ a.s. The law of a normalized GFF on the cylinder σ -algebra is unique (proved), and a normalized GFF exists on a probability space in **Type** (proved, part of **main_result**).

- **GFF plus a (bounded) continuous function.** h is measurable and there is a measurable $f : \Omega \rightarrow C(\mathbb{C}, \mathbb{R})$ on the same probability space with $h - f$ a whole-plane GFF; bounded means each sample $f(\omega)$ is bounded. GM speak of “a coupling”; all conclusions are a.s. statements about h alone, so they transfer to and from an extension of the space.
- h_ε^* and **LFPP**. $h_\varepsilon^*(z) = \lim_n \langle h, p_{\varepsilon^2/2}(z, \cdot) \chi_n \rangle$ with smooth cutoffs χ_n (the heat kernel is not a test function); for a GFF plus a bounded continuous function the limit exists a.s., locally uniformly (proved). D_h^ε is GM’s (1.4): the infimum over piecewise C^1 paths $[0, 1] \rightarrow \mathbb{C}$ of $\int_0^1 e^{\xi h_\varepsilon^*(P(t))} |P'(t)| dt$, the integral taken in $[0, \infty]$.
- α_ε . The *lower* median $\inf\{m : P(X \leq m) \geq 1/2\}$ of $X = D_h^\varepsilon(\{0\} \times [0, 1], \{1\} \times [0, 1])$ under the law of the normalized GFF, following GM l. 223 literally: the *whole-plane* distance between the two sides. DFGPS (l. 285) and DDDF (Thm 1.1(2)) normalize instead by the crossing along paths staying in the unit square, and GM l. 587–590 apply DFGPS Thm 1.2 with GM’s α_ε as if the two agreed. The project proves that along subsequences the ratio of the two medians converges and that the law of the limit’s crossing distance has no gap (so its median is unique, which GM l. 590 also use silently). Hence the limit in Theorem 1.1 changes only by a deterministic positive factor, and Theorem 1.2 is unaffected. The Lean statement keeps GM’s α_ε .
- **Convergence.** “In probability w.r.t. the local uniform topology on $\mathbb{C} \times \mathbb{C}$ ” is: for all $R, \delta > 0$, $P(\sup_{\bar{B}_R(0)^2} |X_\varepsilon - Y| \geq \delta) \rightarrow 0$ along all real $\varepsilon \rightarrow 0^+$ (not a sequence), as in mathlib’s convergence in measure; these events are measurable up to null sets (proved).
- **Determined by h .** Y is a.s. determined by h if $Y = F \circ h$ a.s. for a measurable F . Locality (Axiom II) asks for a measurable F on $\mathcal{D}'(U)$ with $D_h(z, w; U) = F(h|_U)(z, w)$ for all $z, w \in U$ outside one null set; the target carries the product σ -algebra on $\mathbb{C} \rightarrow \mathbb{C} \rightarrow [0, \infty]$, i.e. determination coordinate by coordinate.
- ξ . $\xi = \gamma/d_\gamma$ with $d_\gamma = 2/\chi$ (DG l. 214), and χ is *the* $\chi > 0$ such that, for every zero-boundary GFF on the open unit square and every fixed pair $u \neq v$ in it, a.s. $\log D_{\gamma, \delta}(u, v) / \log \delta^{-1} \rightarrow \chi$, where $D_{\gamma, \delta}$ is DZZ’s Liouville graph distance (DZZ l. 121–122: open balls with rational centres and LQG mass at most δ^2 , covering a path from u to v , paths and balls unrestricted). DZZ Thm 1.1 states two-sided bounds $C^{-1}\delta^{-\chi+\iota} \leq D \leq C\delta^{-\chi-\iota}$ with random C ; for $D \geq 1$ monotone in δ this is the log-limit. Uniqueness and existence of χ are proved on the final route, so ξ is the genuine value.
- **Strong LQG metric.** A measurable $D : \mathcal{D}'(\mathbb{C}) \rightarrow$ continuous metrics with Axioms I–IV whenever h is a GFF plus a continuous function. A continuous metric is a finite, jointly continuous metric on $\mathbb{C}$ whose small balls are Euclidean-small (equivalent to inducing the Euclidean topology), with the compact-open topology and its Borel σ -algebra. Axiom III holds on one null set for all continuous f ; Axiom IV for fixed r, z on a null set depending on them. Probability spaces range over **Type** (universe 0), where every law on a standard Borel space is realized.
- **Theorem 1.1 (S1).** The limit is a random element Y of $C(\mathbb{C} \times \mathbb{C}, \mathbb{R})$ that is a.s. a metric; it is *not* required to be a continuous metric (to induce the Euclidean topology), because GM do not state that in Theorem 1.1.

- **Theorem 1.2 (S4, S5).** Existence: there is a strong γ -LQG metric D such that $\mathfrak{a}_\varepsilon^{-1}D_h^\varepsilon \rightarrow D_h$ in probability for every GFF plus a bounded continuous function (equivalent to “the limit of Theorem 1.1 is a.s. D_h ”, since limits in probability are a.s. unique). Uniqueness: $\exists C > 0$ before the quantifier over (Ω, P, h) , and a.s. $D_h(u, v) = C\tilde{D}_h(u, v)$ for all u, v .

3 Cited results proved in a different form

Each row gives the form proved, its direction relative to the printed statement, and why. In every case the form proved is what GM’s (or the citing paper’s) proof actually uses, and the chain to Theorems 1.1/1.2 is closed with it. “FALSE” means the printed statement has a counterexample; “Lean” that the counterexample is kernel-checked.

CONF L2.10	Only for U = the annulus minus finitely many squares of CONF §3 (the only U at which L2.10 is applied: C:712–742, 1125–1127, 1236–1243, 1331). <i>Weaker domain.</i> For irregular U the white-noise argument (C:722–731) needs the field $h_{t,\infty}$ extended by 0 to be continuous, which can fail. Proved via Berestycki–Powell §1.5.
CONF L2.14	Printed form is FALSE for small n (a disc with a long thin fjord, $n = 1$: the “ ≤ 8 long arcs” dropped at C:855 cost an additive constant) and, as a disconnection statement, trivially true and too weak for L2.15. Proved: $\# \text{good} \geq (1 - AC^{-2} \text{area } U)n - N_0$ with a universal N_0 , in a <i>located</i> form (the disconnecting set within $Cn^{-1/2}$ of ∂U), on Jordan domains. <i>Corrected and stronger.</i> The only consumer (L3.10, C:1584–1590) needs $n_k \geq N_0$ only.
CONF L3.3	The event (3.9)/(3.12) can be empty (pocket configurations). Proved for a grid-component event in its place; L3.6 Step 2 is derived from it. <i>Different event.</i>
CONF L3.6, L3.7	Hypotheses changed: the centre lies on $\partial \mathcal{B}_\tau^\bullet$ a.s. (not surely: unattainable for a metric modified on a null set); conditioning on $\sigma(\mathcal{B}_\tau^\bullet, h _{\mathcal{B}_\tau^\bullet})$ modulo additive constants, with the explicit hypothesis that $\mathcal{B}_\tau^\bullet$ is determined by h modulo constants (the raw forms are false for a GFF with a random additive constant; CONF normalize at C:347, 1154, 1187); radii are stopping times only a.s.; the scales ε are countably valued (Step 3, C:1438, conditions on a countable partition); L3.7’s property A for targets outside $\text{int } \mathcal{B}_\sigma^\bullet$, as used at C:1600. L3.5 with other constants (C:1284’s do not give the bound).
CONF T3.9	Extra hypotheses that CONF use tacitly: $\mathcal{B}_\tau^\bullet$ determined by h modulo constants (C:1431); a complete probability space and a.s. stopping times (C:1302); the arcs lie on Jordan curves $\partial \mathcal{B}_s^\bullet$ (CONF use prime ends at C:835, 1550; the project proves $\partial \mathcal{B}_s^\bullet$ is a Jordan curve); the hit set K is closed and bounded with nonempty interior and Jordan frontier (covers every use, since $z_0 \in \text{int } \mathcal{B}_s^\bullet$); geodesics are leftmost in L2.4’s sense.
DG P3.16	Printed lower bound FALSE (Lean: z, w across a shared grid side; the “largest j' ” at DG:1500–1512 need not exist). Proved with error term $\delta^{1-\zeta} e^{\xi \hat{h}_\delta(v_{S_z})}$ in place of $\delta e^{\xi \hat{h}_\delta(v_{S_z})}$, which is what DG’s own count gives. <i>Weaker.</i> P3.17 absorbs it with a smaller ζ .
DG L3.7	Stated on $[0, 1]^2$ (DG:1096–1102), used on $\mathbb{S}(1/2) = [-1/2, 3/2]^2$ (DG:1729–1731). Proved on $\mathbb{S}(1/2)$ from Ding–Goswami’s coupling. <i>Stronger.</i>
DZZ Cor 3.3	Rate $(\log \delta^{-1})^{-0.1}$ instead of $e^{-(\log \delta^{-1})^{0.9}}$, which does not follow from P3.2. <i>Weaker</i> ; consumers use only that the difference tends to 0.

DZZ P3.2, L3.5	Separate location exponent ξ and diameter exponent $\xi' > \xi$ (Cor 3.9 and L3.10 apply L3.5 at $\hat{\delta} = \delta e^h > \delta$, as DZZ's own proof of P3.2 does at l. 1150); measure confined to $\mathbb{V}$ and clipped covers, since the covering balls of l. 1131–1137 stick out of $\mathbb{V}$; (eq-cell-LQG-compare) only for $S \subseteq \mathbb{V}$ (false at $\partial\mathbb{V}$). Lemma 3.4's event on dyadic box centres (η has no continuous version).
DZZ P3.17	The walled form (Remark 5.2) only for pairs well inside the wall: Remark 5.2 gives no range of pairs and the literal reading fails near the wall. The crude moments (l. 849–857) only for small δ (false as $\delta \rightarrow 1$). <i>Narrower range.</i>
DZZ L5.3, L5.4, L6.1	L5.4/L6.1: segments of length $\delta^{K\iota}$ with $K\chi \geq 2$ (resp. δ^κ) instead of $\delta^{2\iota}$, whose crossing cost $\delta^{-\chi+\iota}$ works only for $\chi \geq 1/2$; the leg of l. 2563 lies in $\mathbb{V}_{u,2\lambda} \not\subseteq \tilde{\mathbb{V}}_{u,v}$, so the leg walls are redefined; the wall family restricted to the central squares DZZ use (translation invariance fails near $\partial\mathbb{V}$). L5.3 part 1: threshold $L^{0.97}$ for $L^{0.95}$ (the errors add up to more), independence gap $L/64$ for $L/25$, near-diagonal pairs excluded, and a rate where l. 2516–2522 say only “with probability tending to 1”. <i>Corrected constants.</i>
DFGPS P4.3	For filled metric balls $\mathcal{B}_s^\bullet$ instead of metric balls: the proof (T:2699–2741) bounds geodesic times $\geq s$ only, and for earlier times needs a point of ∂B_k at distance $\geq s$ from 0, which fails when B_k surrounds a hole. GM use P4.3 only for filled balls. <i>Different object.</i>
DFGPS L3.5, P3.1	U connected (the proof uses it, T:1580; false otherwise). <i>Stronger hypothesis.</i> L3.20 with $u \neq v$ (false for $u = v$); P3.18/L3.20/L3.22 at scales $\varepsilon(3/4)^k$ with mesh $\tau\mathfrak{r}/40$ (the printed mesh does not fit).
DFGPS L2.14	Only the existence of cluster points $\mathfrak{c}_r \in (0, \infty)$ of $r\mathfrak{a}_{\varepsilon/r}/\mathfrak{a}_\varepsilon$ with the Λ -bounds, not a limit (§4). <i>Weaker</i> ; L2.13, its only consumer, needs no more.
DDDF Prop 29, Thm 1(2)	On $(-1, 2)^2$, the domain where they are used. Constants: L25 drops a factor 2 (condition $(C_1 + C_2)^2 + 2(C_1 + C_2)C_3 \leq C_3^2$); L9 needs $\alpha = s/\sigma^2$, not $s/(2\sigma^2)$; (6.99) does not cover $n = 0$.
LM L2.1	The harmonic clause of the Markov decomposition holds a.s. only (the every- ω form is unprovable: fields on null sets). <i>Weaker.</i>
MQ L4.1	For general radii $\rho_1 < \rho_2 < 1$ instead of $7r/8, 15r/16$: GM L2.7 needs the comparison on B_1 inside B_{1+s} ; MQ's proof does not use the ratios. <i>Stronger.</i>
GM L3.9	U bounded and w within $\varepsilon^{1+\nu}\mathfrak{r}/100$ of x : FALSE for unbounded U and near $\partial(\mathfrak{r}U)$. <i>Weaker</i> , as used.
GM L4.18	Off by one ($K = m = 1, \alpha = 0.99, \delta = 0.98: 0.9604 > e^{-1.92}$); two corrected forms proved, both give the decay L4.21 needs.
GM L5.6, L5.8	The tube $V_r(z)$ of l. 2945 need not contain $z \pm 2r$; geometry restated.
GM §2.4, §3.3, §4	The tightness facts S2.4a–e, used without proof, are proved (S2.4e from DFGPS T1.5 and LM L3.1); the footnote at l. 1222 is repaired (P3.4 with $C_1 = (C' + C_*)/2$); $\lambda_4 > 1$ is required (l. 2555 says $\lambda_4 \geq 1$; the only application has $\lambda_4 = 4$); the proof of L4.14 is repaired (statement unchanged).

4 Proofs that follow a different route

- **DFGPS L2.14, L2.17, Thm 1.2.** L2.14's proof (T:1080–1098) needs joint convergence of $(h, \mathfrak{a}_\varepsilon^{-1}D_h^\varepsilon)$ and the median of the internal crossing of $[0, 1]^2$ for the *limit* metric (not a function of the limit, by DFGPS's remark after L2.11); the cluster-point form is proved and suffices. In L2.17 Step 4 (T:1268–1278) the passage from conditioning on $h|_V$ to $h|_{\bar{V}}$ does not follow from the facts available (a two-fair-bits model satisfies them and fails the conclusion); the lemma is proved on DFGPS's route with $\sigma(h|_{\bar{V}})$ carried from the LFPP stage. In the Thm 1.2 assembly,

joint convergence in law is stated in countably many coordinates of h (mathlib has no Prokhorov theory on $\mathcal{D}'$), and the glued metric is built from *truncated* local limits $\min(d(a, b), d(a, \partial W))$, which are determined by the limit metric; the untruncated local limit need not be.

- **LM Thm 1.7.** Step 3 (l. 1075–1084) bounds the length of a near-geodesic of $D(\cdot, \cdot; V)$ inside a square S by $\sup_S D + \varepsilon^2$, but the replacement path must stay in V , and the needed bound is false in general (Jordan V with fjords at all scales). Proved with $U = \mathbb{C}$, $V = B_n$, LM’s Steps 1–2 verbatim, Step 3 replaced by a product bound with an auxiliary near-geodesic, and the variance estimate in LM’s own conditional form (l. 1011–1026).
- **DZZ L3.12.** The split of $\mathcal{C}_{i,\text{cross}}$ into two segments (l. 1471–1477) presumes that the enclosure is a closed ring. The ring is obtained from the four-crossing form of L3.16 (DZZ’s duality argument, l. 1009–1011, gives it) and L3.12 consumes the cyclic ring.
- **DZZ L5.3 part 1.** Restructured into four nodes on DZZ’s own boxes $\mathcal{E}^*$ (l. 2363–2378), with the η -chaos as the local proxy and near-pair cut-offs, around the gaps listed in the table.
- **Identification of χ .** DZZ Thm 1.1 is obtained from Prop 5.1 with explicit rates, Borel–Cantelli along dyadic δ and monotonicity between dyadics, a step DZZ do not write; the frozen definition (balls and paths unrestricted) is connected to DZZ’s confined distance by the first-exit sandwich in Prop 5.1’s proof.
- **CONF T3.9.** CONF treat the arc hit sets $\mathcal{I}_k$ (C:1559–1568) as measurable; a priori they are Σ_2^1 . They are replaced, up to null sets, by GM’s analytic arc relation (GM l. 1705–1708).

5 Global conventions

- **Almost-sure versions.** Sure statements that cannot hold for a metric or field modified on a null set are stated a.s.: boundary points of filled balls, stopping times, the harmonic part of the Markov decomposition, geodesic selectors. Where an argument needs a complete probability space (CONF T3.9), the result is transferred back to a general P by completion.
- **σ -algebras modulo additive constants** in CONF §3, because a whole-plane GFF in this formalization may carry a random additive constant; the raw σ -algebras would leak it. GM’s §4 keeps its raw σ -algebras and receives the normalized form.
- **Explicit constants and parameter ranges** wherever the paper says “sufficiently small/large” ($\lambda_4 > 1$, $\varepsilon_0 > 0$, $\nu s < 1$, $\delta < 1/8$, the thresholds of DZZ §5). Constants that depend only on the law are chosen before the probability space, as GM’s Thm 4.2 needs.
- **Measurability made explicit:** hypotheses such as $\sigma(\mathcal{B}_\tau^\bullet, h|_{\mathcal{B}_\tau^\bullet}) \leq \mathcal{F}$, analytic sets for geodesic events, outer measure for events not obviously measurable, the product σ -algebra in Axiom II. Definitions are total: limits that may fail, empty infima and the fallback $\chi = 1/2$ get junk values, shown never to be hit on the final route.

6 Where the details are

Repository github.com/leobon12/lean-research, folder `formalizations/lqg_metric`: `DEVIATIONS.md` (every departure from a source, 362 entries with line numbers), `DECISIONS.md` and `decisions/` (the decisions behind them), `ERRATA.md` (errors found in each source paper, with the fix used), `STATEMENT_SPEC.md` and `FOUNDATIONS.md` (the frozen statements and definitions), `audits/`, and `reports/PAPERS-*.md` (the Lean entry point of each cited result).